Java Programming 3

Week 7 - JDBC - JdbcClient - Profiles

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Agenda this week

JDBC
Implementing the repository
JdbcClient
Spring profiles

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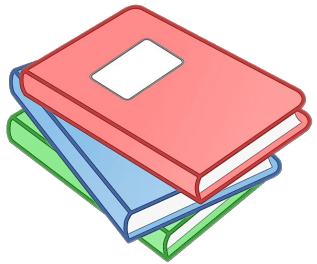
Tutorials

- JDBC: http://tutorials.jenkov.com/jdbc/index.html
- JDBC and Spring: https://www.baeldung.com/spring-6-jdbcclient-api
- Spring JDBC Reference: https://docs.spring.io/spring-framework/reference/data-access/jdbc.html
- Spring Profiles: https://www.baeldung.com/spring-profiles
- JdbcClient: https://www.baeldung.com/spring-6-jdbcclient-api



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Agenda this week

JDBC
Implementing the repository
JdbcClient
Spring profiles

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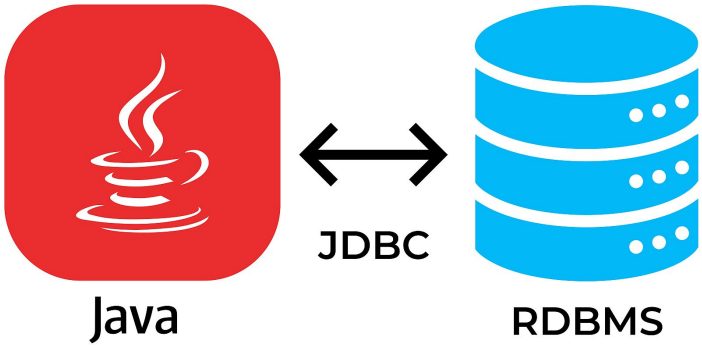
JDBC

- API that allows a Java program to communicate with a relational database
- Independent of the vendor
- Access using SQL Queries
- Part of standard Java (java.sql.*)



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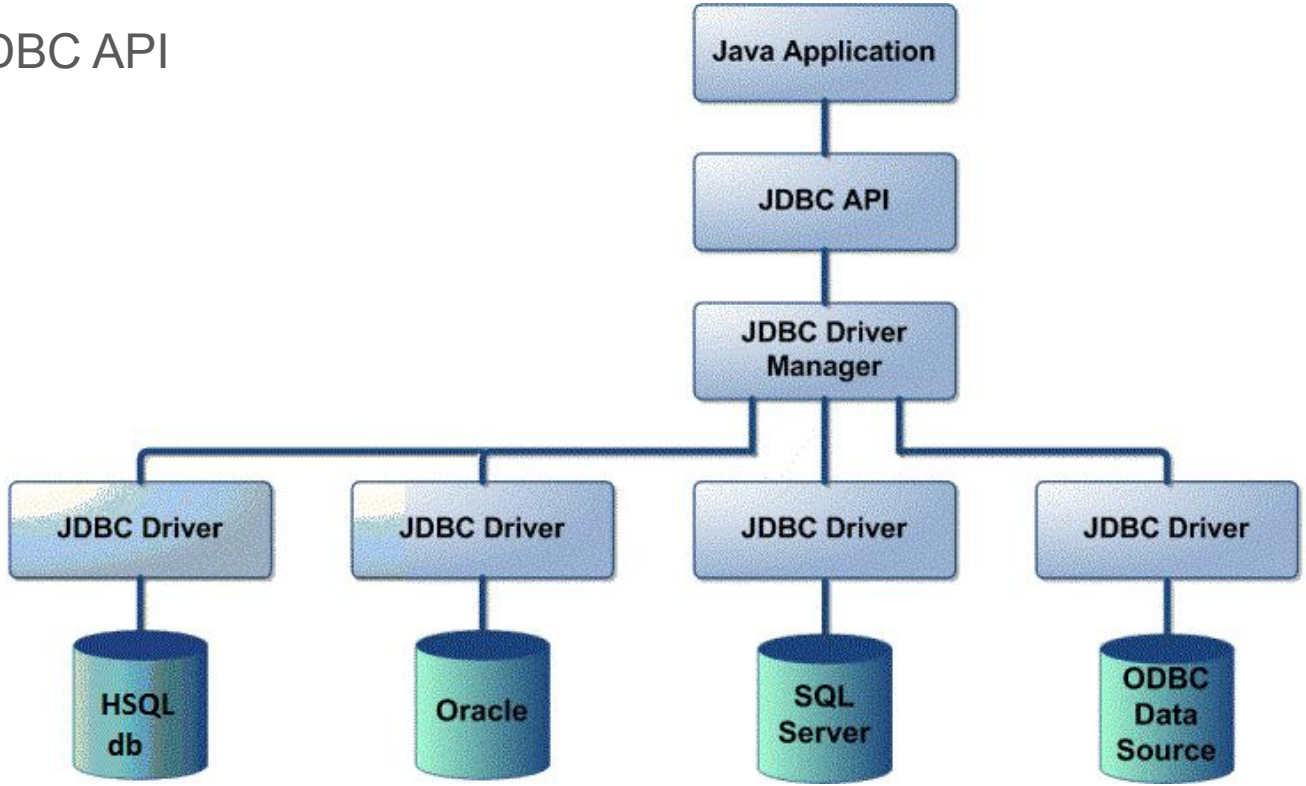
JDBC Driver

- Implements the JDBC API
- Vendor specific



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H2 Database

- We will use H2 DB in the examples
  - Lightweight, small, fast, in memory
  - Perfect for development and testing
  - Not suitable for production: you will use PostgreSQL instead
- We will have to add the driver to the gradle dependency!

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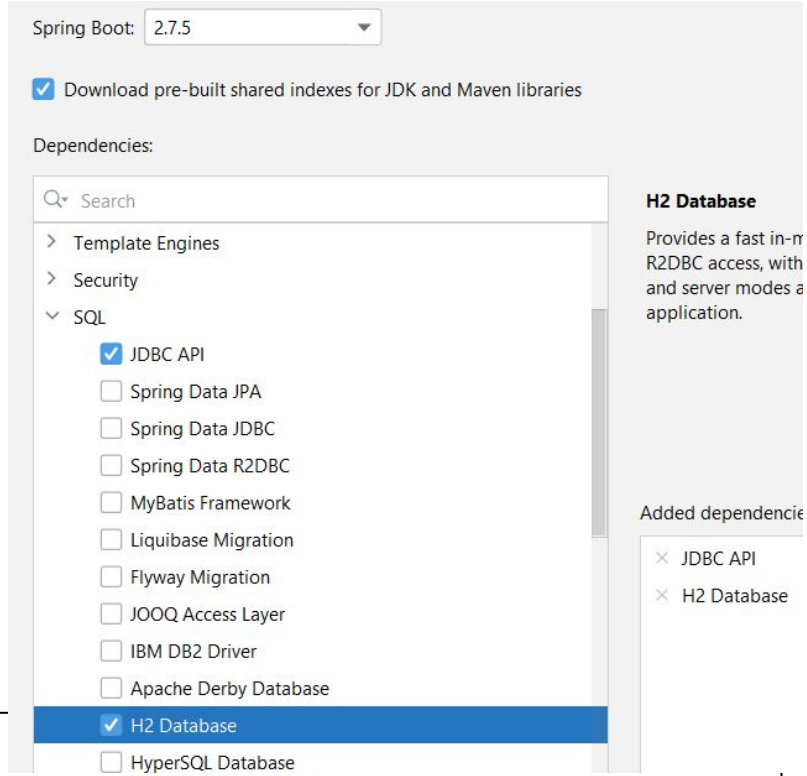
Create new Spring project

- Add 2 dependencies
  - JDBC API
  - H2 Database



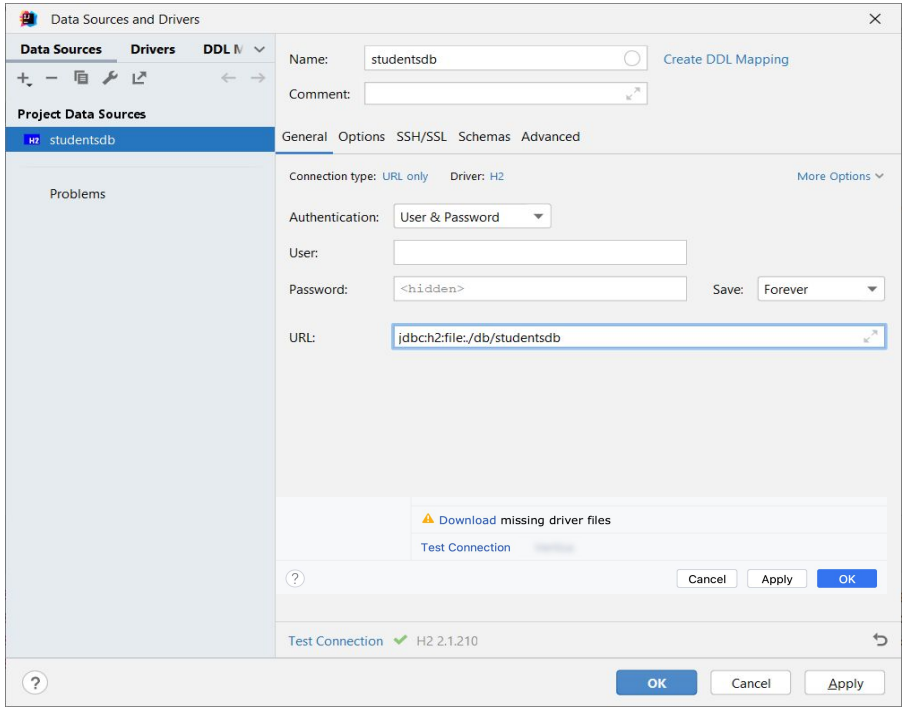

```
dependencies {
    //…
    implementation ("org.springframework.boot:spring-boot-starter-jdbc")
    runtimeOnly ("com.h2database:h2")
}
```

Create DB first: IntelliJ Database View

- Connection Type: URL Only
- URL: **jdbc:h2:file:./db/studentsdb**

  → Creates DB in ./db folder

  - If intellij proposes to download missing driver files, do so.



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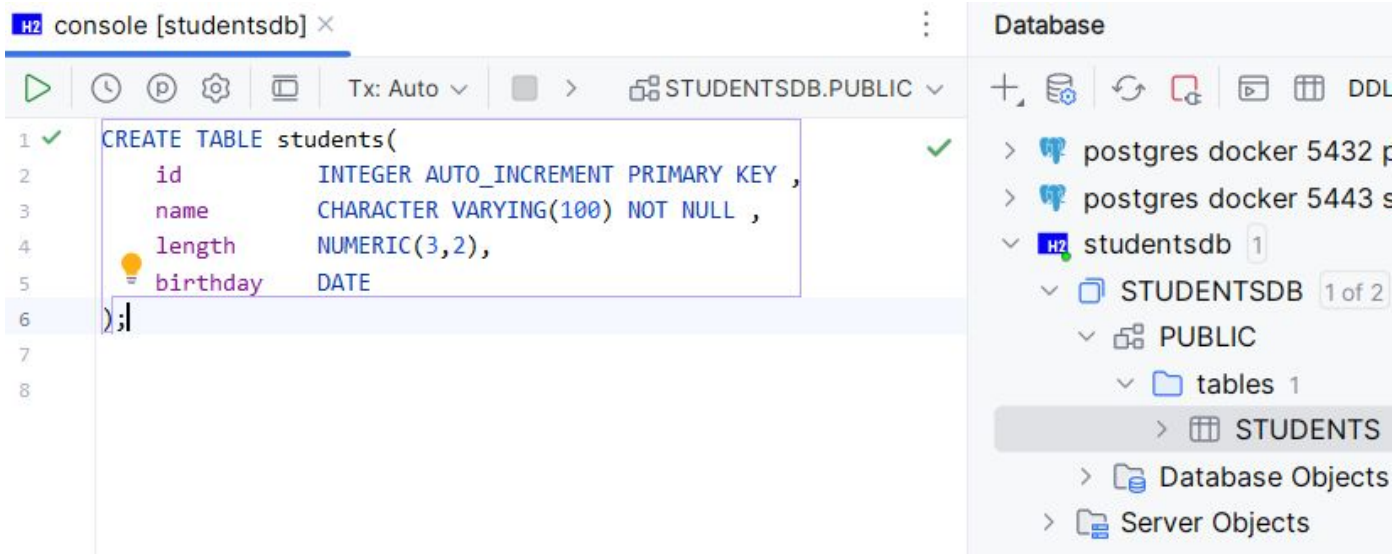
Create table STUDENT

- Use the DB View
- Create columns id (PK, auto_increment), name, LENGTH, BIRTHDAY



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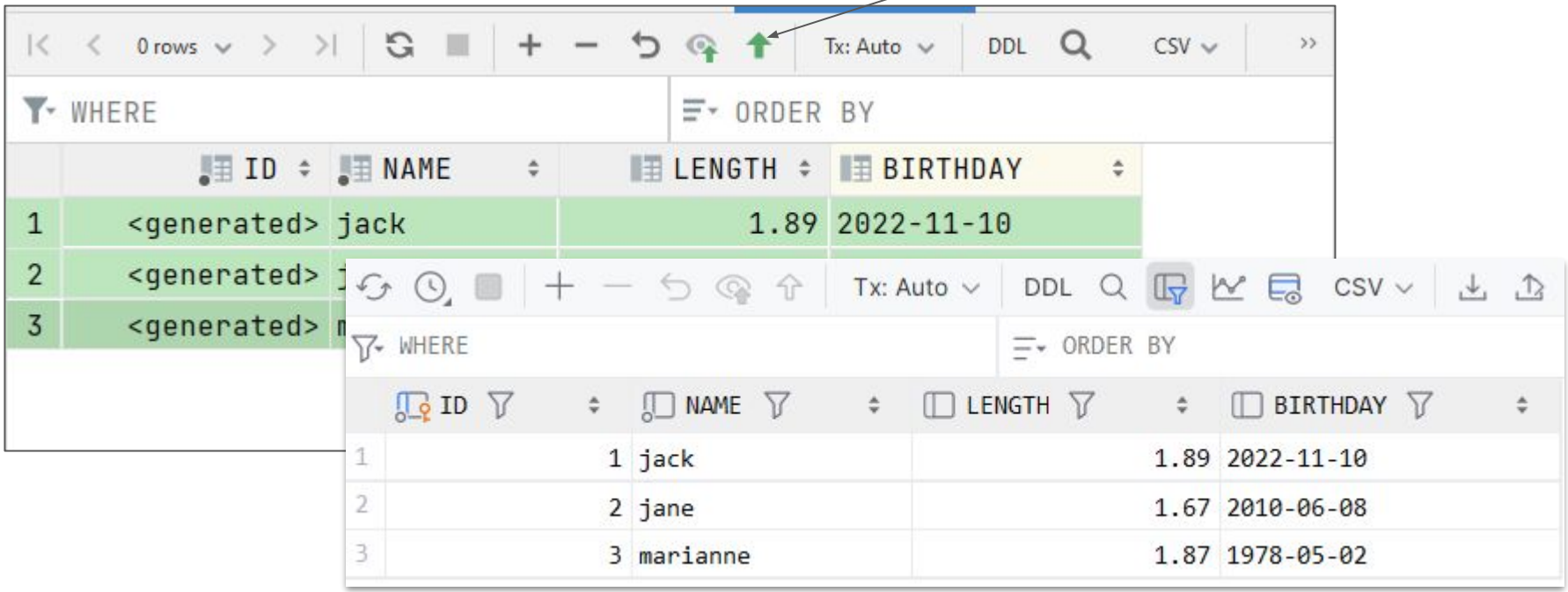
Add some records…



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JDBC: basic steps

1. Add JDBC Driver
2. Create a `Connection` to the database
3. Create a `Statement` (or `PreparedStatement`)
4. Perform query
5. Process the result (`ResultSet`)
6. Close `ResultSet` - `Statement` - `Connection`

Slides code

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Connection

- Create a connection to the Database via **`DriverManager.getConnection(..)`**
  - You can provide username and password as extra parameters

```
@Component
public class JdbcQuery implements CommandLineRunner {
 @Override
 public void run(String... args) throws Exception {
  Connection connection = DriverManager.getConnection("jdbc:h2:file:./db/studentsdb");
  Statement statement = connection.createStatement();
  ResultSet resultSet = statement.executeQuery("SELECT * FROM students");
  while (resultSet.next()) {
   System.out.printf("%d %s\n",resultSet.getInt("id"), resultSet.getString("name"));
  }
  resultSet.close();
  statement.close();
  connection.close();
 }
}
```

```
1 jack
2 jane
3 marianne
```

```
A file can only be opened by one process, deactivate the
database in your IntelliJ tool window before running this.
(This problem does not occur with network databases)
```

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Closing the connection

- Close the connection if it is no longer used!
  - Typically the DB can only serve a limited number of connections at the same time…
- Use try (*with resources)* to be sure → will automatically close connection
  - You can open multiple resources in one try (with resources)

```
public void run(String... args) throws Exception {
 try (Connection connection = DriverManager.getConnection("jdbc:h2:file:./db/studentsdb");
      Statement statement = connection.createStatement()) {
  try (ResultSet resultSet = statement.executeQuery("SELECT * FROM students")) {
   while (resultSet.next()) {
    System.out.printf("%d %s\n",resultSet.getInt("id"), resultSet.getString("name"));
   }
  }
 }
}
```

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Statement

- You create a Statement to perform a query on the database
  - You can create multiple Statements on a connection
- Close the Statement after use (eg using try with resources)!

```
public void run(String... args) throws Exception {
 try (Connection connection = DriverManager.getConnection("jdbc:h2:file:./db/studentsdb");
      Statement statement = connection.createStatement()) {
  try (ResultSet resultSet = statement.executeQuery("SELECT * FROM students")) {
   while (resultSet.next()) {
    System.out.printf("%d %s\n",resultSet.getInt("id"), resultSet.getString("name"));
   }
  }
 }
}
```

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ResultSet

- The executeQuery method returns a ResultSet. Via the ResultSet you can retrieve the rows returned by the query…
  - you can execute multiple queries **sequentially** on a statement
- Close the ResultSet after use: you can only have one ResultSet open on a statement.

```
public void run(String... args) throws Exception {
 try (Connection connection = DriverManager.getConnection("jdbc:h2:file:./db/studentsdb");
      Statement statement = connection.createStatement()) {
  try (ResultSet resultSet = statement.executeQuery("SELECT * FROM students")) {
   while (resultSet.next()) {
    System.out.printf("%d %s\n",resultSet.getInt("id"), resultSet.getString("name"));
   }
  }
 }
}
```

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ResultSet

- The next() method moves the cursor to the next row
  - It returns false if there is none

```
public void run(String... args) throws Exception {
 try (Connection connection = DriverManager.getConnection("jdbc:h2:file:./db/studentsdb");
      Statement statement = connection.createStatement()) {
  try (ResultSet resultSet = statement.executeQuery("SELECT * FROM students")) {
   while (resultSet.next()) {
    System.out.printf("%d %s\n",resultSet.getInt("id"), resultSet.getString("name"));
   }
  }
 }
}
```

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ResultSet

- On a row, you can retrieve values of different columns using the column name or the column index (starts from **1**)
  - Using name is preferred

```
try (Connection connection = DriverManager.getConnection("jdbc:h2:file:./db/studentsdb");
    Statement statement = connection.createStatement()) {
 try (ResultSet resultSet = statement.executeQuery("SELECT * FROM students")) {
  while (resultSet.next()) {
   System.out.printf("%d %s\n",resultSet.getInt("id"), resultSet.getString("name"));
   System.out.printf("%d %s\n",resultSet.getInt(1), resultSet.getString(2));
  }
 }
}
```

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ResultSet get…: what SQL types?

- getString(...): for character based types (char, varchar, varchar2, …)
  - Works on any type (does a toString)
- getInt(...), getLong(...): integer types (smallint, int, …)
- getFloat(...), getDouble(...): decimal types (decimal, real, double precision, …)
- getBoolean(...): boolean types
- getDate(...): for dates
  - Returns java.sql.Date object → convert it to LocalDate for use in Java…
- more…

```
Date birthDay = resultSet.getDate("BIRTHDAY");
LocalDate localDate = birthDay.toLocalDate();
```

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Insert, delete or update data?

```
try (Connection connection = DriverManager.getConnection("jdbc:h2:file:./db/studentsdb")) {
 try (Statement statement = connection.createStatement()) {
  int result = statement.executeUpdate("""
   INSERT
   INTO students(name, length, birthday)
   VALUES('AN',1.65,'1967-3-12')
   """);
  System.out.println(result  + " row(s) inserted");

  result = statement.executeUpdate("""
   DELETE
   FROM students
   WHERE name = 'AN'
   """);
  System.out.println(result + " row(s) deleted");
 }
}
```

- Use executeUpdate method for updates (inserts, deletes)
- Returns number of rows affected (1 in the above cases)

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Exercise:

- Create new Spring application
  - Add JDBC and H2 dependencies
- Use the Database view in IntelliJ to:
  - Create table students: a user has an id, name, length and birthday
  - Add some data
- Write a small application (commandlinerunner) that shows all students sorted by birthday
- Perform some updates and deletes on the data
- What happens if you keep the connection open on the IntelliJ Database tool?

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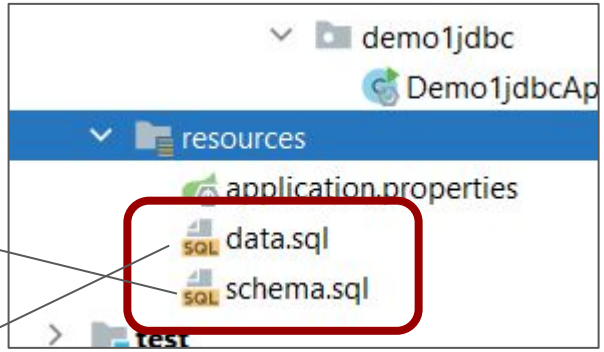
Creating the database tables and loading initial data?

- You can do this in database tool, or using Java code
- Or: you can add a `schema.sql` and `data.sql` to the resources folder
- Spring will run the .sql files once when the application starts...

```sql
DROP TABLE IF EXISTS persons; -- not needed for in memory DB

CREATE TABLE persons(
   id INTEGER AUTO_INCREMENT PRIMARY KEY,
   name CHARACTER VARYING(100) NOT NULL,
   firstname CHARACTER VARYING(50)  NOT NULL,
   remark CHARACTER VARYING(256)
);
```

```sql
INSERT INTO persons(name, firstname,remark)
VALUES ('JONES', 'JACK','of all trades'),
      ('POTTER', 'JACK','Lilly''s dad'),
      ('POTTER', 'MIA','Lilly''s mum'),
      ('REED', 'JACK','union');
```



Configure the datasource in Spring...

- Spring needs to know where to connect: you can add this info in the `application.properties`

```
spring.datasource.url=jdbc:h2:mem:personsdb
spring.datasource.username=sa
spring.datasource.password=
spring.sql.init.mode=always
```

We use a H2 **memory** database now: it will only exist in memory. Ideal for developing and testing…

If not using a memory database Spring does not run the schema.sql and data.sql, you need to add this last line...

Slides code: persons_datasource DataSourceRunner

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Exercise:

- Create a small Spring application (no Spring MVC, just a commandlinerunner) that loads 10 students (name, length, birthday) into a H2 **memory** database at startup:
  - Configure in the application.properties
  - use schema.sql and data.sql to create the table and load the data
- The application asks for the name via the console
- The application shows all records that match the name
- Use this query: `"SELECT * FROM students WHERE name = '" + name + "'"`

```
Enter the name of the student:jack
Found following info for student jack:
id:1, name: "jack", length: 1.92, birthday: 10/11/2022
Enter the name of the student:jef
Found following info for student jef:
Enter the name of the student:matthijs
Found following info for student matthijs:
id:2, name: "matthijs", length: 1.67, birthday: 08/06/2010
Enter the name of the student:_
```

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Exercise:

- Create a small Spring application (no Spring MVC, just a commandlinerunner) that loads 10 students (name, length, birthday) into a H2 memory database at startup:
  - Configure in the application.properties
  - use schema.sql and data.sql to create the table and load the data
- The application asks for the name via the console
- The application shows all records that match the name
- Use this query: `"SELECT * FROM students WHERE name = '" + name + "'"`

→ Try using `JACK' OR '1'='1` as name, what happens?

→ Example of SQL Injection!

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PreparedStatement

The values are inserted at the ? in the PreparedStatement.
The prepared statement can execute the SAME statement multiple times with DIFFERENT parameters

- You can *prepare* a SQL statement:
  - The SQL will be precompiled in the DB → you can prepare it beforehand
  - You can change the parameters
  - It is a good protection against SQL injection…
  - You als get rid of the mix of double and single quotes in your statemen

```
try (PreparedStatement statement = connection.prepareStatement("INSERT INTO students(name, LENGTH, BIRTHDAY)
VALUES(?,?,?)")) {
    for (int i=0;i<10;i++) {
        statement.setString(1, "AN" + i);
        statement.setDouble(2, 1.78);
        statement.setDate(3, Date.valueOf(LocalDate.of(1987, 1 + i, 23)));
        int result = statement.executeUpdate();
        System.out.println(result);
    }
}
```

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Slides code: preparedStatement

Exercise:

- Use a PreparedStatement in the previous exercise: does it solve the SQL Injection problem?

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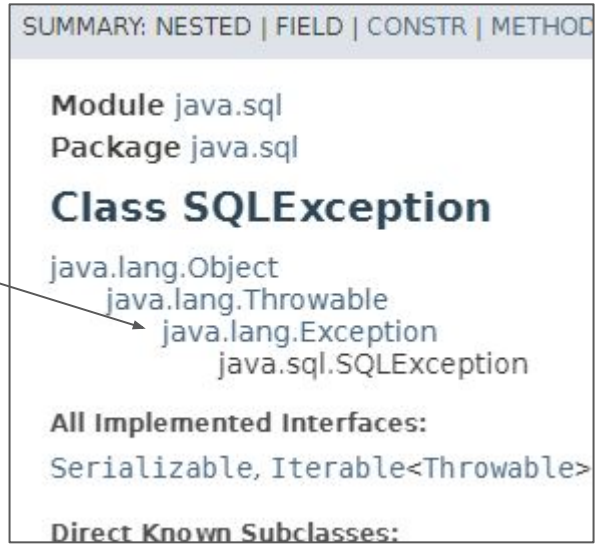
SQLExceptions

- If something goes wrong, JDBC throws an SQLException
  - Example: DB offline, wrong table, column does not exist, …
  - It is a *checked* exception: you have to write exception handling code for it…
  - Not very fine-grained: you get an SQLException for all kinds of problems

Java API Documentation: SQLException
inherits from Exception
so it is a **checked** exception…



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Transactions in JDBC

- Sometimes you need to perform more than one query together.
- Example: transfer money to other account
  - Query one: remove money from the first account
  - Query two: add money to the second account
    - If for some reason the second query does not succeed, the first one should roll back!
- In a relational database you use transactions for this
- You can also perform transactions using JDBC

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Transactions in JDBC

```
try (Connection connection = DriverManager.getConnection("jdbc:h2:mem:personsdb","sa","")) {
 connection.setAutoCommit(false);
 try (Statement statement = connection.createStatement()) {
  try {
   statement.executeUpdate("""
      INSERT
      INTO persons(name, firstname, remark)
      VALUES('Truus','Trampoline','Lastly through a hogshead of real fire!')
    """);
   if (new Random().nextBoolean()) throw new SQLException("Problem!");
   statement.executeUpdate("DELETE FROM persons WHERE firstname LIKE '%ER'");
   System.out.println("No problem, inserting and deleting...");
   connection.commit();
  } catch (SQLException e) {
   System.out.println("Problem, rolling back insert!");
   connection.rollback();
  }
}
```

We mimic an SQLException here, just for the demo. If the exception is thrown, the transaction will rollback INSERT will never be executed without DELETE!

Slides_code: persons_datasource TransactionRunner

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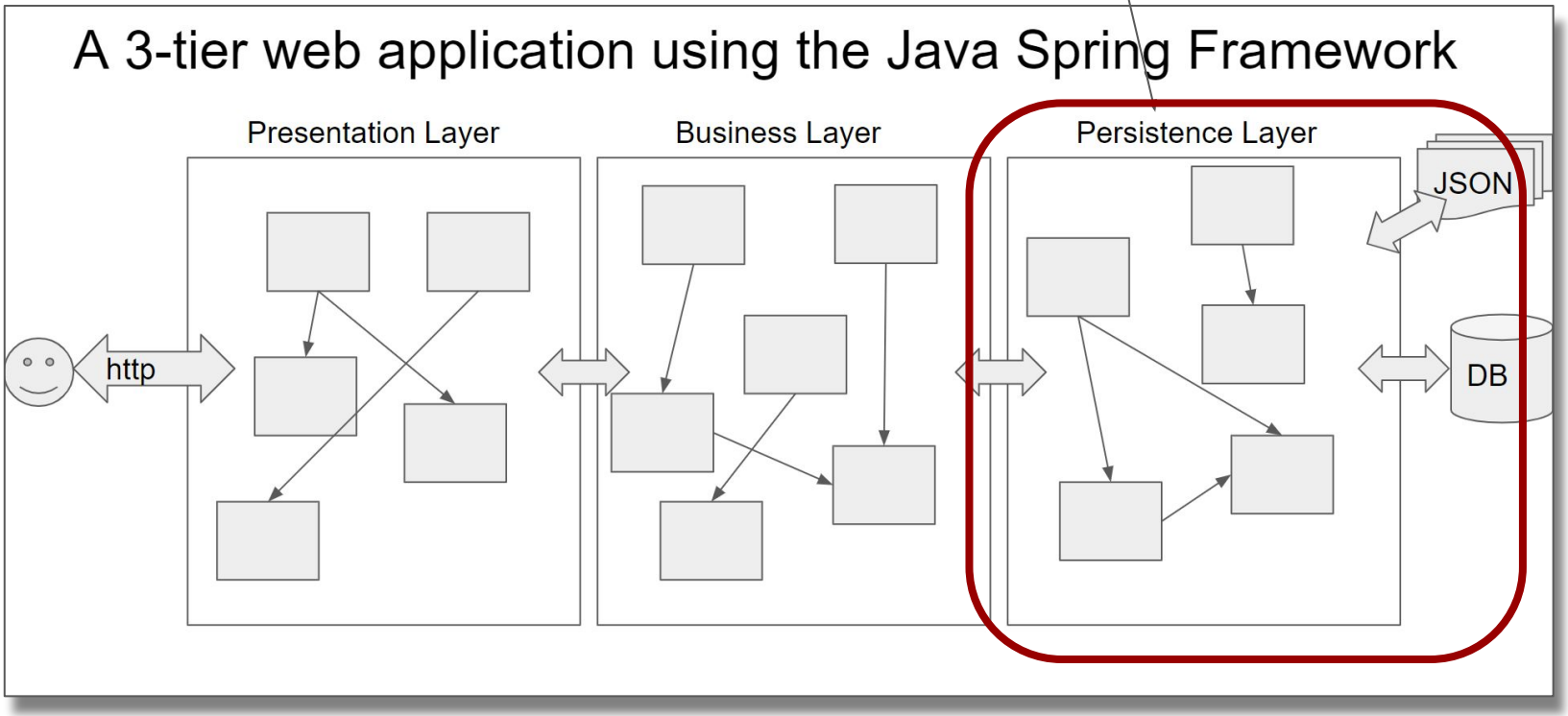
The persistence layer

We will use JDBC in the Persistence Layer.
We will create Repository classes for it...



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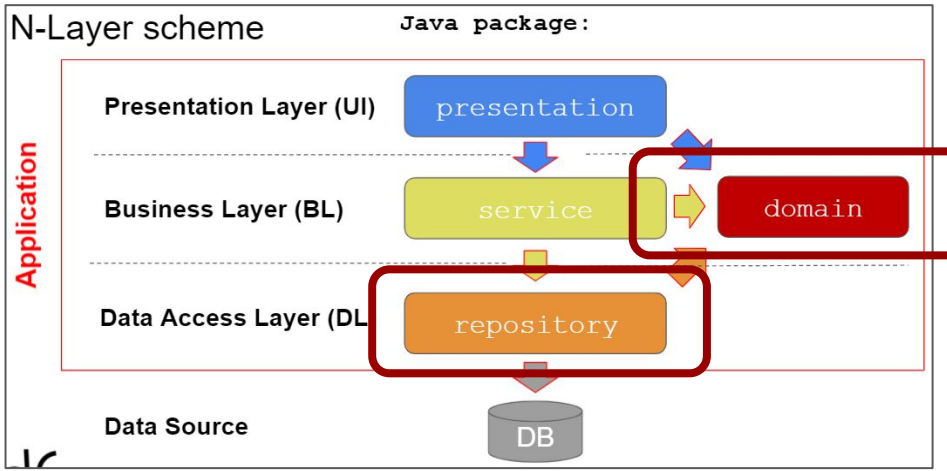
Remember: Domain and Repository

- Domain classes map to database tables (“entities”)
- They have an id to contain the PK of the record
- We create Repository classes for each entity.
- Annotate with @Repository (= a Spring Component)
- The Repository classes will contain the JDBC code
- Repository has methods to
  - Query entities (findBy…)
  - Create entities (create…)
  - Update entities (update…)
  - Delete entities (delete…)



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Exercise: create the StudentJdbcRepository(1/6)

- Create the domain class Student (id, name, length, birthday)
- Configure your project with a database with some Students
  - Use application.properties - schema.sql - data.sql
- Create the StudentRepository interface:

```
public interface StudentRepository {
   List<Student> findAll();
   Student createStudent(Student student);
   void updateStudent(Student student);
   void deleteStudent(int id);
}
```

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Exercise: create the StudentJdbcRepository(2/6)

- Create the StudentJdbcRepository(implements StudentRepository)
- Implement the findAll() method
- You need the database URL, username and password to create the connection → values from the application.properties files can be retrieved as follows:

```
public StudentJdbcRepository(@Value("${spring.datasource.url}") String dbURL,
 @Value("${spring.datasource.username}") String user,
 @Value("${spring.datasource.password}:''") String password) {
 this.dbURL = dbURL;
 this.user = user;
 this.password = password;
}
```

:'' default value
(if property is not found)
is an empty string

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Exercise: create the StudentJdbcRepository(3/6)

- Create the presentation class StudentMenu
  - The StudentRepository is injected (*we will skip the service layer for this exercise…*)
  - We will show a small console menu like this:
- Test findAll!

```
Welcome to the Student Management System
========================================
1) List students
2) Add student
3) Update student
4) Delete student
Make a choice:
```

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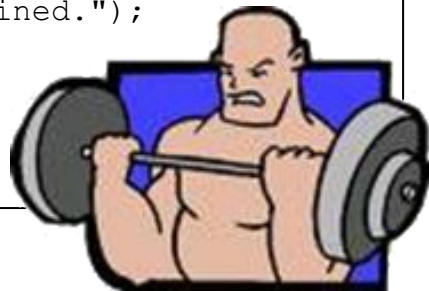
Exercise: create the StudentJdbcRepository(4/6)

- Implement the createStudent() method
- You need the id of the newly created Student. How?

```
try (PreparedStatement statement
  = connection.prepareStatement("INSERT INTO students(..)", Statement.RETURN_GENERATED_KEYS)){
   //…
   int result = statement.executeUpdate();
   if (result != 0) {
       try (ResultSet generatedKeys = statement.getGeneratedKeys()) {
           if (generatedKeys.next()) {
               createdStudent.setId(generatedKeys.getInt(1));
           }
           else {
               throw new SQLException("Creating student failed, no id obtained.");
           }
       }
   }
}
```



Exercise: create the StudentJdbcRepository(5/6)

- Implement the updateStudent() and deleteStudent() methods
- Example run:

```
Welcome to the Student Management System
========================================
1) List students
2) Add student
3) Update student
4) Delete student
Make a choice:3
Student{id=1, name='jane', length=1.87, birthday=2010-06-08}
Student{id=2, name='marianne', length=1.23, birthday=1978-05-02}
Student{id=3, name='Truus', length=2.0, birthday=1954-03-12}
Student{id=4, name='An', length=1.67, birthday=1987-02-23}
Which student (id)?4
new name:Jef
new length:1.87
new birthday (mm-dd-yyyy):02-23-1987
```

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Exercise: create the StudentJdbcRepository(6/6)

- What about those SQLExceptions?
- We don’t want the repository to throw SQLExceptions to the higher layers
  - Create an unchecked exception class DatabaseException
  - Wrap the SQLException in it and throw to higher layers
  - Catch in the presentation layer and show a message…

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Agenda this week

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JdbcClient: removes the boilerplate code

```java
@Component
public class JdbcClientRunner implements CommandLineRunner {
 private JdbcClient jdbcClient;

 public JdbcClientRunner(JdbcClient jdbcClient) {
   this.jdbcClient = jdbcClient;
 }

 @Override
 public void run(String... args) throws Exception {
  //Query the database
  jdbcClient.sql("SELECT * FROM persons")
    .query((RowCallbackHandler) rs ->
       System.out.println(rs.getString("id") + " " + rs.getString("name")));
 }
}
```

Spring provides JdbcClient.
It handles connections, statement, closing, exceptions etc behind the scenes, taking parameters (url, user, password) from application.properties

RowCallbackHandler is executed for each resultset row

Old JDBC Code:

```java
public void run(String... args) throws Exception {
   try (Connection connection = DriverManager.getConnection("jdbc:h2:mem:peronsdb","sa","")){
       try (Statement statement = connection.createStatement()) {
           try (ResultSet resultSet = statement.executeQuery("SELECT * FROM students")) {
               while (resultSet.next()) {
                   System.out.println(resultSet.getString("id") + " " + resultSet.getString("name"));
               }
           }
```

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Example: convert ResultSet to List of domain objects

```
public List<Person> findByName(String name){
    return jdbcClient.sql("SELECT * FROM PERSONS WHERE NAME = ?")
     .param(name)
     .query((rs, rowNum) -> new Person(rs.getInt("id"),
      rs.getString("name"),
      rs.getString("firstname"),
      rs.getString("remark")))
     .list();
}
```

A RowMapper implementation (here a lambda). It provides code to convert the ResultSet into a domain Object. JdbcClient will apply it to all records in the ResultSet!

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More on JdbcClient

- `For updates use the update() method`
- Passing parameters
  - param can have an optional first parameter with the number or the name of the parameter
  - you can chain multiple param(...) calls
  - you can pass multiple parameters to the params(...) call
    - params can also take a map with name/value pairs
- `JdbcClient internally uses a PreparedSatement`

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Example: query for a single object

```
public Person findById(int id) {
 return jdbcClient.sql("SELECT * FROM persons WHERE id = :personid")
   .param("personid",id)
   .query(this::mapRow)
   .single();
}

private Person mapRow(ResultSet rs, int i) throws SQLException {
 return new Person(rs.getInt("id"),
    rs.getString("name"),
    rs.getString("firstname"),
    rs.getString("remark"));
}
```

named parameter

The mapRow method reference does the same as the lambda in the previous example, but can be reused in other queries returning Person(s)

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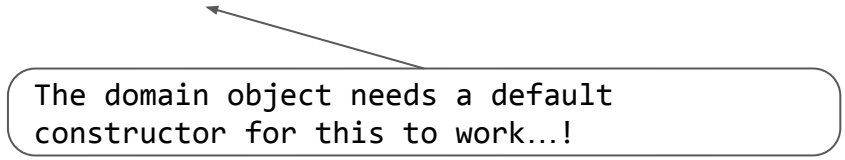
The RowMapper

- You can pass a `RowMapper` to the `query` method. It maps a row in the `ResultSet` to the Domain object. It can be done in different ways:
  - Implement a `RowMapper` (using class, a lambda expression, referencing a method from the repository…)
  - Pass the Domain class to generate the `RowMapper` based on the Domain object



Slides code: persons_datasource RowMapperRunner

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Example: RowMapper from domain class

```
public List<Person> findByFirstName(String firstName) {
 return
  jdbcClient.sql("SELECT * FROM persons WHERE firstname = ?")
    .param(firstName)
    .query(Person.class)
    .list();
}
```

- Works only if column names match setter names

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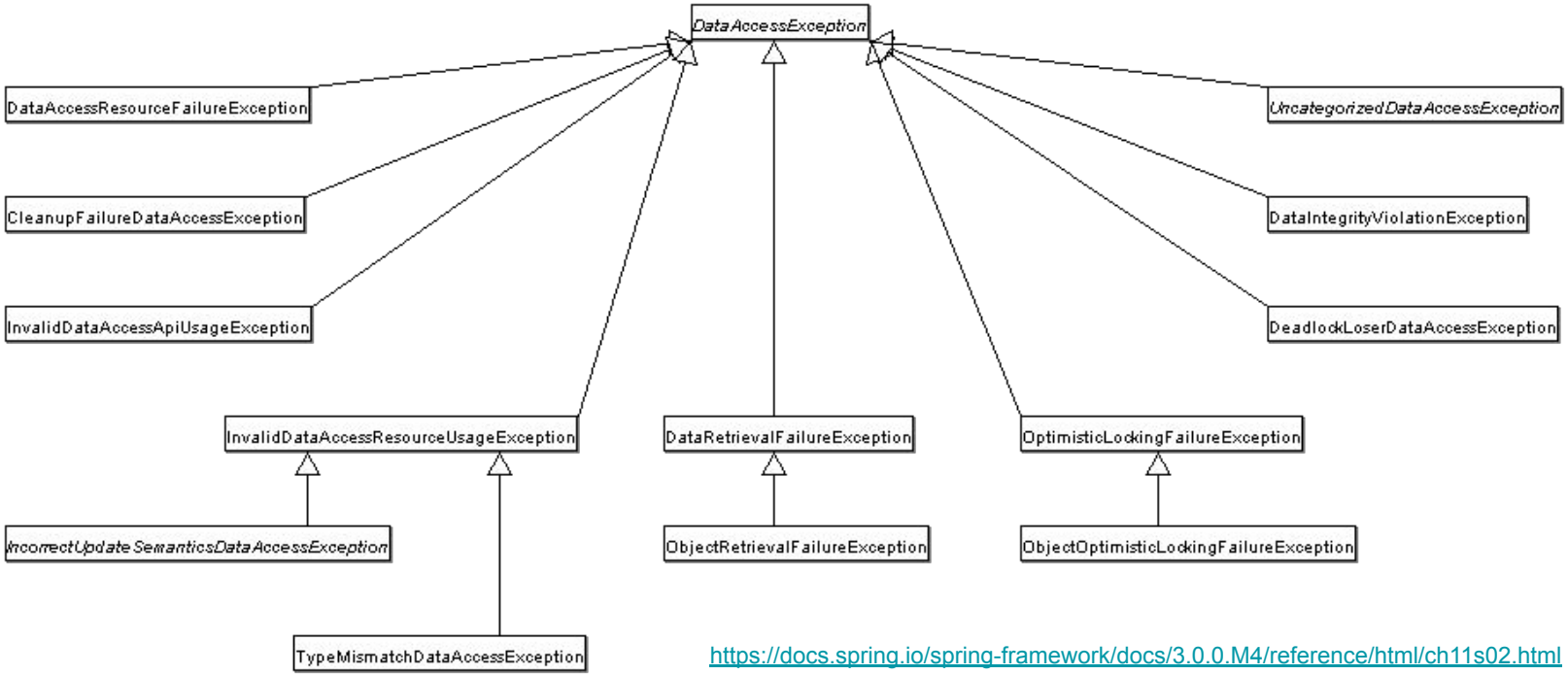
And the SQLException?

Spring provides more fine-grained exceptions so you can write better exception handling.
The exceptions are unchecked: you don't need to handle them!

- In a @Repository it is replaced by a hierarchy of `DataAccessExceptions`



https://docs.spring.io/spring-framework/docs/3.0.0.M4/reference/html/ch11s02.html

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Saving entities to the database

- How is the retrieving of the id implemented?

```
public Person save(Person person) {
 JdbcClient.update("INSERT INTO persons (name, firstname, remark) VALUES (?, ?, ?)",
  person.getName(),
  person.getFirstName(),
  person.getRemark());
 // set Person id??
 return person;
}
```

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Using a KeyHolder

```
public Person save(Person person) {
 KeyHolder keyholder = new GeneratedKeyHolder();
 jdbcClient.sql("""
   INSERT INTO persons(name, firstname,remark)
    VALUES (:name, :firstname,:remark)
   """)
  .params(Map.of(
      "name", person.getName(),
      "firstname", person.getFirstName(),
      "remark", person.getRemark()))
  .update(keyholder);
 person.setId(keyholder.getKey().intValue());
 return person;
}
```

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Or using a SimpleJdbcInsert object

```
private SimpleJdbcInsert personInserter;
private JdbcClient jdbcClient;

public PersonJdbcClientRepository(JdbcClient jdbcClient, DataSource dataSource) {
 this.jdbcClient = JdbcClient;
 personInserter = new SimpleJdbcInsert(dataSource).withTableName("persons")
  .usingGeneratedKeyColumns("id");
}
//...
public Person save(Person person) {
 int personId = personInserter.executeAndReturnKey(Map.of(
  "name", person.getName(),
  "firstname", person.getFirstName(),
  "remark", person.getRemark())
 ).intValue();
 person.setId(personId);
 return person;
}
```

JdbcTemplate

- JdbcClient is a wrapper around Spring's JdbcTemplate
  - JdbcClient it has a simpler fluent API
  - JdbcClient is available from Spring Framework 6.1/Spring Boot 3.2
  - JdbcTemplate calls the standard JDBC API
- for complex cases/older Spring versions your may want to use the underlying JdbcTemplate
- You can find some JdbcTemplate examples in `Slides code: persons_datasource JdbcTemplateRunner`

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Exercise:

- Create a second implementation of the StudentRepository, this time using JdbcClient!
  - Inject the JdbcClient
  - Use a private mapRow method to map a ResultSet entry to a Student
  - Use SimpleJdbcInsert to retrieve the id when creating a new Student
  - Think about exception handling…
- Test this implementation…

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Agenda this week

JDBC
Implementing the repository
JdbcClient
Spring profiles

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Spring profiles

- https://www.baeldung.com/spring-profiles
- If you like to use another DB technology for development versus testing or production?
  - Spring provides the possibility to define *profiles*
    - You can select the active profile(s) in `application.properties`:
      - `spring.profiles.active=prod`
      - `spring.profiles.active=prod,JdbcTemplate`
    - You can create separate `application-dev.properties`, `application-prod.properties`, …
  - In your code: using the `@Profile` annotation you can map beans to different profiles

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Example: H2 for development - PostgreSQL for production

- Add `org.postgresql:postgresql` dependency is to build.gradle.kts
- Put PostgreSql attributes into application-prod.properties

```
spring.datasource.url=jdbc:postgresql:personsdb
spring.datasource.username=postgres
spring.datasource.password=Student_1234
spring.sql.init.mode=always
spring.sql.init.schema-locations=classpath:schema-prod.sql
spring.sql.init.data-locations=classpath:data-prod.sql
```

Normally you will not reinitialise your production db, and only use the schema and data files for development.

- PostgreSql dialect may differ

```
DROP TABLE IF EXISTS persons;
CREATE TABLE persons(
   id  SERIAL PRIMARY KEY,
   name CHARACTER VARYING(100) NOT NULL,
   firstname CHARACTER VARYING(50)  NOT NULL,
   remark CHARACTER VARYING(256)
);
```

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Slides code: profiles

Or: Configure the database from code with @Profile

- Annotating beans with @Profile(“dev”) and @Profile(“prod”), will only use them if these profiles are active
- Create two @Configuration classes H2DatabaseConfig and PostgreSqlDatabaseConfig

dev

```
@Bean
public DataSource dataSource(){
   DataSource dataSource = DataSourceBuilder
.create()
        .driverClassName("org.h2.Driver")
            .url("jdbc:h2:mem:studentdb")
            .username("sa")
            .password("")
            .build();
   return dataSource;
}
```

prod

```
@Bean
public DataSource dataSource(){
   DataSource dataSource = DataSourceBuilder
.create()

.driverClassName("org.postgresql.Driver")
            .url("jdbc:postgresql:pro3_db")
            .username("postgres")
            .password("Student_1234")
            .build();
   return dataSource;
}
```

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Slides code: profiles_code

Configuring @Profile from code

- Add `H2LoadData` and `PostgresLoadData` class (annotate with `@Component` and `@Profile`)
- Add `@PostConstruct` method in each class:

dev

```
@PostConstruct
public void loadData() {
 jdbcClient.sql("DROP TABLE IF EXISTS PERSONS").update();
 jdbcClient.sql("""
    CREATE TABLE PERSONS(
       ID INTEGER AUTO_INCREMENT PRIMARY KEY,
       NAME CHARACTER VARYING(100) NOT NULL,
       FIRSTNAME CHARACTER VARYING(50)  NOT NULL,
       REMARK CHARACTER VARYING(256)
    )
       """).update();
 jdbcClient.sql("""
  INSERT INTO PERSONS(NAME, FIRSTNAME,REMARK)
 VALUES ('JONES', 'JACK','of all trades'),
       ('POTTER', 'JACK','Lilly''s dad'),
       ('POTTER', 'MIA','Lilly''s mum'),
       ('REED', 'JACK','union');
       """).update();
}
```

prod

```
@PostConstruct
public void loadData() {
 jdbcClient.sql("DROP TABLE IF EXISTS PERSONS").update();
 jdbcClient.sql("""
  CREATE TABLE PERSONS(
  ID  SERIAL PRIMARY KEY,
  NAME CHARACTER VARYING(100) NOT NULL,
  FIRSTNAME CHARACTER VARYING(50)  NOT NULL,
  REMARK CHARACTER VARYING(256)
  );
  """).update();
 jdbcClient.sql("""
  INSERT
  INTO PERSONS(NAME, FIRSTNAME,REMARK)
   VALUES ('JONES', 'JACK','of all trades'),
       ('POTTER', 'JACK','Lilly''s dad'),
       ('POTTER', 'MIA','Lilly''s mum'),
       ('REED', 'JACK','union');
       """).update();
}
```

Profiles: there’s more…

- Profiles can be configured in many other ways and can be used for many other purposes, we only look at a small example
- For more info, check the tutorial: https://www.baeldung.com/spring-profiles

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Exercise: using profiles

- Install the PostgreSQL database system on your platform
- Create a database using PostgreSQL tooling
- Try to connect using the IntelliJ Database tooling
- Walk through the previous slides and try to create 2 profiles, switch between the two and test your application.
    - Try to configure the databases using 2 application.properties files
    - Try to configure the databases from code using the @Profile annotation

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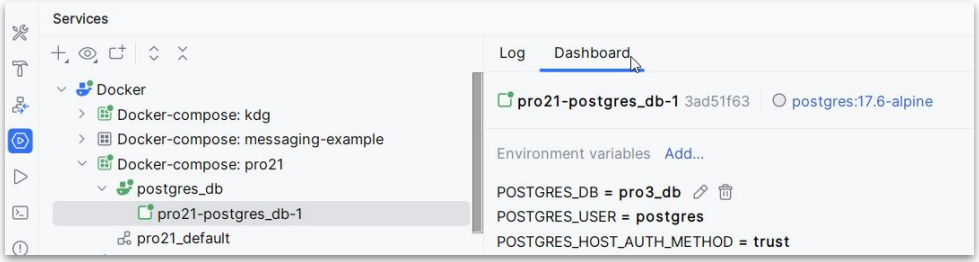
Exercise: using profiles (continued)

If you are familiar with using Docker, you can add a docker-compose.yml file to your project root directory. Be sure the docker engine is running and start the postgresql database from within Intellij!
In the directory with the docker-compose.yml run
docker compose up -d



docker-compose.yml

```yaml
version: '3.8'
services:
 postgres db:
   image: postgres:17.6-alpine
   restart: always
   environment:
     POSTGRES DB: 'pro3 db'
     POSTGRES USER: 'postgres'
     POSTGRES_PASSWORD: 'Student_1234'
   ports:
     - '5444:5432'
   volumes:
      - ./postgres/data:/var/lib/postgresql/data
```

Uses port 5444 when running from docker to avoid conflicts with a possible local installation of postgres

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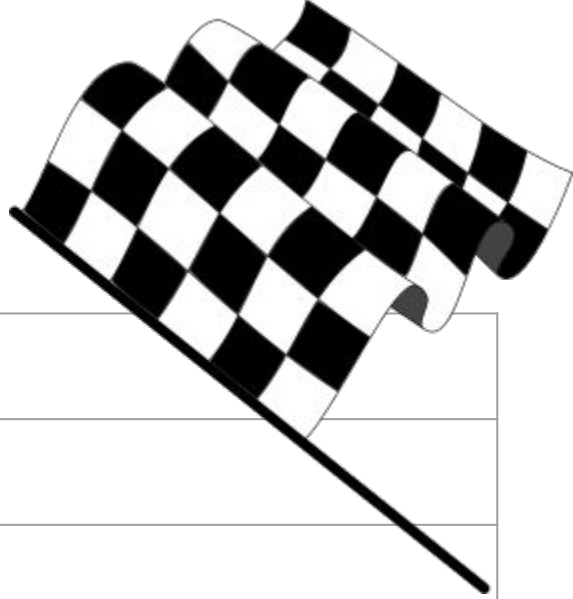
Agenda this week



JDBC
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Spring profiles

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Project

- Implement the Repository of your 3 entities using Spring JDBC (JdbcClients) in combination with a H2 database
- Use a schema.sql and data.sql to load initial data.
- Use profiles to be able to switch between the old implementation (using Java Collections) and the new implementation (using JDBC)

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